

The Kraus Formalism in Quantum Computing

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Introduction

The **Kraus formalism** provides the most general mathematical description of quantum evolution for open quantum systems.

It describes:

- Decoherence
- Dissipation
- Quantum noise
- Measurements
- Environmental interactions

It generalizes ordinary unitary quantum mechanics to realistic quantum devices interacting with an environment.

Closed Quantum Systems

For a closed quantum system:

$$|\psi\rangle \rightarrow U|\psi\rangle,$$

where U is unitary:

$$U^\dagger U = I.$$

For density matrices:

$$\rho \rightarrow U\rho U^\dagger.$$

This evolution preserves:

- Probability
- Purity
- Reversibility

Open Quantum Systems

Realistic quantum systems interact with:

- Thermal environments
- Measurement devices
- Electromagnetic noise
- Imperfect hardware

The evolution becomes:

- Non-unitary
- Irreversible
- Probabilistic

Examples:

- Decoherence
- Relaxation
- Dephasing
- Spontaneous emission

The Kraus Representation

The most general physical quantum evolution is

$$\rho' = \sum_k K_k \rho K_k^\dagger,$$

where K_k are called:

- Kraus operators
- Operation elements
- Noise operators

This defines a quantum channel:

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger.$$

Trace Preservation

Physical states must satisfy:

$$\text{Tr}(\rho) = 1.$$

The Kraus operators obey the completeness relation:

$$\sum_k K_k^\dagger K_k = I.$$

This guarantees:

$$\text{Tr}(\rho') = 1.$$

The map is therefore trace preserving.

System + Environment Picture

Consider:

System + Environment.

The combined system evolves unitarily:

$$U_{SE}.$$

The subsystem evolution is obtained by tracing out the environment:

$$\rho'_S = \text{Tr}_E \left[U_{SE}(\rho_S \otimes \rho_E) U_{SE}^\dagger \right].$$

Open-system dynamics emerge naturally from entanglement with the environment.

Derivation of Kraus Operators

Suppose the environment initially starts in:

$$|0_E\rangle.$$

The Kraus operators are:

$$K_k = \langle k_E | U_{SE} | 0_E \rangle.$$

The environment basis states:

$$|k_E\rangle$$

represent different environmental outcomes.

Bit-Flip Noise Channel

Suppose a qubit flips with probability p :

$$|0\rangle \leftrightarrow |1\rangle.$$

The Kraus operators are:

$$K_0 = \sqrt{1-p} I,$$

$$K_1 = \sqrt{p} X.$$

The Pauli X -matrix is:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Bit-Flip Channel Evolution

The density matrix evolves as:

$$\rho' = (1 - p)\rho + pX\rho X.$$

Interpretation:

- With probability $1 - p$: nothing happens.
- With probability p : the qubit flips.

This is a simple quantum noise model.

Amplitude Damping Channel

Amplitude damping models spontaneous emission:

$$|1\rangle \rightarrow |0\rangle.$$

The Kraus operators are:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix},$$

$$K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}.$$

The parameter γ controls the decay probability.

Physical Meaning of Amplitude Damping

Amplitude damping produces:

- Energy loss
- Relaxation
- Irreversible decay
- Decoherence

Examples:

- Spontaneous photon emission
- Superconducting qubit relaxation
- Atomic decay

A quantum channel must be:

- 1 Linear
- 2 Completely positive
- 3 Trace preserving

Such maps are called:

CPTP maps.

The Kraus representation gives the most general CPTP map.

Complete Positivity

Positivity alone is insufficient in quantum mechanics.

If the system is entangled with an ancilla, the map:

$$\mathcal{E} \otimes I$$

must remain positive.

This is called:

Complete positivity.

It is essential for:

- Entanglement
- Multipartite systems
- Quantum computing

Connection to Lindblad Dynamics

The Kraus formalism describes:

- Finite-time evolution

Continuous-time evolution is described by the Lindblad master equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_i \gamma_i \left(L_i \rho L_i^\dagger - \frac{1}{2} \{L_i^\dagger L_i, \rho\} \right).$$

Kraus operators can be derived from short-time Lindblad evolution.

Quantum Measurements

Generalized measurements are described by:

$$M_k = K_k^\dagger K_k.$$

The probability of outcome k is:

$$p_k = \text{Tr}(K_k \rho K_k^\dagger).$$

The post-measurement state becomes:

$$\rho_k = \frac{K_k \rho K_k^\dagger}{p_k}.$$

The operators:

$$M_k = K_k^\dagger K_k$$

form a Positive Operator-Valued Measure (POVM).

POVMs generalize projective measurements.

Applications:

- Weak measurements
- Noisy measurements
- Quantum tomography
- Quantum communication

Bloch Sphere Interpretation

Unitary evolution:

- Rotates the Bloch sphere

Kraus evolution can:

- Shrink the sphere
- Distort the sphere
- Shift the sphere

Examples:

- Depolarization: shrink toward center
- Amplitude damping: drift toward $|0\rangle$

Applications in Quantum Computing

The Kraus formalism is fundamental for:

- Quantum error correction
- Noise-aware quantum circuits
- Quantum communication
- Open-system quantum simulation
- Quantum sensing
- Quantum machine learning
- NISQ device modeling

It is the standard language of realistic quantum hardware.

Quantum Error Correction

Noise channels can be represented as:

$$\rho' = \sum_k K_k \rho K_k^\dagger.$$

Quantum error correction attempts to reverse these errors.

Examples:

- Bit-flip correction
- Phase-flip correction
- Surface codes
- Stabilizer codes

Summary

The Kraus formalism provides the most general description of open quantum evolution.

Key ideas:

- Quantum channels
- CPTP maps
- Decoherence
- Dissipation
- Generalized measurements

It forms a central foundation of:

- Quantum computing
- Quantum information
- Quantum optics
- Quantum sensing
- Quantum technologies